

Class 26: Dimension and Rank

Definition: If V is any vector space, the dimension of V is

4 Important facts:

1. If $\dim V = n$, any set of more than n vectors in V will be linearly dependent.
2. If V has one basis with n vectors, then every basis for V will have n vectors.
3. If S is any finite spanning set for a nonzero vector space V , then there is some subset of S that is a basis for V .
4. Any linearly independent set of vectors in V can be extended to form a basis for V .

Example $V = \left\{ \begin{bmatrix} a - 2b + 5c \\ 2a + 5b - 8c \\ -a - 4b + 7c \\ 3a + b + c \end{bmatrix} \mid a, b, c \in \mathbb{R} \right\}$ Find a basis for V .

Definition: The set of all linear combinations of the rows of a matrix A is called _____ denoted by _____.

1.

2.

Example: Find two different bases for the row space of the matrix A .

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 1 & -2 & 3 \\ -1 & 0 & -3 & 2 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \\ -1 & -2 & -3 \\ 4 & 3 & 2 \end{bmatrix}$$

We can now use the row reduced form of matrix A to find bases for the row space, the column space, and the null space of A !

What is the dimension for each of these spaces?

$$\dim \text{Nul } A =$$

$$\dim \text{Col } A =$$

$$\dim \text{Row } A =$$

Four Fundamental Spaces:

There are four important spaces associated with any $m \times n$ matrix A .

Big idea for future work:

Think about it questions

#6 If A is 7×5 and has rank 2, find the dimension of $\text{Nul } A$ and $\text{Row } A$ and the rank of A^T .

#8 If A is 6×8 and has four pivot columns, what is the dimension of $\text{Nul } A$? Is $\text{Col } A = \mathbf{R}^n$?

#20 Suppose a nonhomogeneous system of six linear equations in eight unknowns has a solution, with two free variables. Is it possible to change some constants on the equations' right sides to make the new system inconsistent?

#22 Is it possible that all solutions of a homogeneous system of ten equations in twelve variables are multiples of one fixed nonzero solution?

23 A homogeneous system of twelve equations in eight unknowns has two fixed solutions that are not multiples of each other, and all other solutions are linear combinations of these two solutions. Can the set of all solutions be described with fewer than twelve homogeneous equations?